NASA: Lunar Survival Activity

Objective: Students will analyze and think critically about items necessary for their survival in space.

Time Needed: 20-30 minutes

Background Knowledge Required: NONE

Teacher Prep:

Print copies of next page for students. (1 copy per student/1 copy per pair of students if you want them to work in pairs or small groups.)

Teaching Instructions:

Read the instructions on the student page for the students

Give the students 5-10 minutes to rank their items

Go through the NASA rankings (below). Have students write down the NASA rankings on their page.

To score: Mark the number of points that your score differs from the NASA ranking, then add up all of the points. Disregard plus or minus differences. The lower the total, the better the score.

0-25	**Excellent**
26-32	**Good**
33-45	**Average**
46-55	**Fair**
56- 70	**Poor**
71 – 112	**Very Poor**

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Scenario:

You are a member of a space crew originally scheduled to meet with the mothership on the moon. However, due to mechanical failures, your ship was forced to land at a location 200 miles from the meeting point.

During reentry and landing, much of the equipment on the ship was damaged. Below are the 15 items that were left intact and undamaged after the landing. Your task is to rank them in order of importance for your crew's 200-mile trip to the mothership. Number 1 is the most important item and 15 is the least important. The items on the list below are in no particular order.

Your Ranking	Item	NASA Ranking
	Box of matches	
	Food concentrate	
	50 feet of nylon rope	
	Parachute silk	
	Portable heating unit	
	Two .45 Caliber pistols	
	One case of dehydrated milk	
	Two 100 lb. tanks of oxygen	
	Stellar Map	
	Self-inflating life raft	
	Magnetic compass	
	20 liters of water	
	Signal flares	
	First aid kit, including injection needle	
	Solar powered FM receiver-transmitter	

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Answer Sheet

Your Ranking	Item	NASA Ranking
	Box of matches	**15**
	Food concentrate	**4**
	50 feet of nylon rope	**6**
	Parachute silk	**8**
	Portable heating unit	**13**
	Two .45 Caliber pistols	**11**
	One case of dehydrated milk	**12**
	Two 100 lb. tanks of oxygen	**1**
	Stellar Map	**3**
	Self-inflating life raft	**9**
	Magnetic compass	**14**
	20 liters of water	**2**
	Signal flares	**10**
	First aid kit, including injection needle	**7**
	Solar powered FM receiver-transmitter	**5**

#15 Box of matches – worthless, no oxygen on the moon
#4 Food concentrate – efficient means of supplying energy
#6 Nylon rope – useful for climbing cliffs and tying the injured together
#8 Parachute silk – protection from the sun
#13 Portable heating unit – not necessary due to the sun's heat (unless you happen to be on the dark side of the moon)
#11 Pistols – possible method of self-protection
#12 Dehydrated milk – similar to food concentrate
#1 Oxygen tanks – without oxygen, there is no survival in space
#3 Stellar Map – Navigation
#9 Self inflating life raft – CO_2 bottle might be used for self-propulsion
#14 Magnetic compass – useless, since there's not a polarized magnetic field on the moon
#2 20 liters of water – like oxygen, necessary for survival
#10 Signal flares – for use as a distress signal when the mothership is in sight
#7 First Aid Kit – for aiding the injured
#5 FM receiver-transmitter – for communication with the mothership